Scalability & Future Plan

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user authentication; trips are visible to all visitors	Privacy and data ownership	High
2	Free-tier backend sleeps when idle	Slow first request	Medium
3	AI output is not checked against real places or prices	Possible inaccurate suggestions	Medium
4	No rate limiting on the AI endpoint	Cost and abuse risk	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single free instance on Render	Paid instances with autoscaling; caching of popular destinations
2	Data Storage	Supabase free tier, single shared table	Per-user rows with row level security; indexes; larger plan
3	Performance	One blocking LLM call per request	Response streaming, background jobs and caching of repeated requests
4	Security	Secrets in env variables, open access	Supabase Auth, row level security, rate limiting, CORS limited to the frontend domain

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User login, per-user trips, regenerate a single day	Next 1-2 months	Personal data and a more flexible planner
Phase 3	Group trips with combined preferences and shareable trip links	3-4 months	Covers group planning scenarios
Phase 4	Maps and real place data, destination photos, budget tracking, PDF export	6 months	More accurate and more useful itineraries